

Notes: Berger, Milbradt, Tourre, and Vavra (AER: Insights, 2025)

Optimal Mortgage Refinancing with Inattention

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1 Big picture

- **Question.** How should fixed-rate mortgage borrowers optimally refinance when refinancing has fixed costs and borrowers pay attention to rates only sporadically?
- **Main friction.** Borrowers cannot continuously monitor mortgage rates. Attention arrives at Poisson times with intensity λ .
- **Core result.** Inattention lowers the optimal refinancing rate-gap threshold. When inattentive borrowers finally pay attention, they should refinance at smaller rate gaps than a perfectly attentive borrower.
- **Why this matters.** The earlier ADL benchmark assumes perfect attention and therefore classifies many observed decisions as mistakes. With inattention, many of these decisions are optimal.
- **Two apparent mistakes.** Inattention can rationalize both errors of omission, refinancing too slowly, and some errors of commission, refinancing at rate gaps that look too low under perfect attention.
- **Analytical contribution.** Under Brownian mortgage rates, the paper derives closed-form refinancing thresholds and ergodic distributions of rate gaps.
- **Sufficient statistic.** The difference between the average rate gap upon prepayment and the average rate gap depends only on mortgage-rate volatility σ , moving/amortization rate ν , and attention rate λ .
- **Empirical estimate.** Using Fannie Mae SFLP microdata from 1999–2023, the preferred attention rate is $\lambda = 23.4\%$ per year when $\sigma = 0.70\%$.
- **Quantitative implication.** Inattention reduces average refinancing thresholds from about 1.09% to 0.75% when $\sigma = 0.70\%$.
- **Economic takeaway.** Mortgage refinancing mistakes are partly a benchmark problem: borrowers who seem too slow or too quick under perfect attention can be close to optimal once inattention is modeled.

2 Data and measurement

2.1 Mortgage contract and rate gap

- Borrowers hold a long-term fixed-rate prepayable mortgage with coupon c_t and unit balance.
- The current market mortgage rate is m_t .
- The rate gap is

$$z_t = c_t - m_t.$$

- A positive rate gap means the borrower can lower the mortgage coupon by refinancing.
- Refinancing requires a fixed cost ψ .

Interpretation: The rate gap is the key state variable. A large positive gap creates refinancing gains, but fixed costs and attention frictions delay refinancing.

2.2 Attention process

- Borrowers pay attention only at random decision times.
- Attention arrives according to a Poisson process with intensity λ .
- A higher λ means more frequent checking.
- Perfect attention is the limiting case $\lambda \rightarrow \infty$.

- Complete inattention is the limiting case $\lambda \rightarrow 0$.

Interpretation: The borrower may pass through profitable refinancing opportunities without acting because the opportunity is not observed.

2.3 Moving or amortization rate

- The mortgage exits for exogenous reasons at rate ν .
- This can represent moving, refinancing for unrelated reasons, or exponential amortization.
- In the empirical implementation, the paper uses $\nu = 11.33\%$ per year.

Interpretation: A borrower who expects the mortgage to disappear soon has less reason to pay a large fixed cost to refinance.

2.4 Empirical mortgage data

- The empirical sample is a random sample of 780,000 thirty-year fixed-rate mortgages.
- Mortgages are issued between January 1999 and September 2023.
- The panel contains 35.42 million mortgage-month observations.
- The paper estimates a refinancing rate for each mortgage-month using Fannie Mae SFLP data and borrower/loan covariates.
- The average rate gap is 0.15%; the average rate gap upon prepayment is 0.78%.

Interpretation: These two empirical rate-gap moments pin down the attention rate through the paper's sufficient statistic.

3 Models and empirical specifications

3.1 General setup

The mortgage rate follows a diffusion:

$$dm_t = \mu(m_t)dt + \sigma(m_t)dW_t,$$

with infinitesimal generator $\mathcal{L}$. The borrower minimizes discounted mortgage costs, including interest payments and fixed refinancing costs.

The optimal refinancing rule is a threshold rule:

$$\text{refinance at attention time if } c_t - m_t \geq \theta(m_t).$$

Interpretation: The borrower does not refinance every time the rate gap is positive. The gap must be high enough to compensate for the fixed cost and the option value of waiting.

3.2 Threshold indifference condition

At the threshold, the borrower is indifferent between refinancing and staying put:

$$V(m, m) + \psi = V(m, m + \theta(m)). \tag{1}$$

Interpretation: The left side is the cost of refinancing: pay ψ and reset the coupon to m . The right side is staying with a coupon $m + \theta(m)$.

3.3 Hamilton–Jacobi–Bellman equation

The value function satisfies

$$(\rho + \nu + \lambda)V(m, c) = c + \nu V(m, m) + \lambda \min\{V(m, c), V(m, m) + \psi\} + \mathcal{L}V(m, c). \quad (2)$$

Interpretation: The HJB has four forces: interest costs, exogenous mortgage exit, attention/refinancing opportunities, and mortgage-rate dynamics.

3.4 General comparative static

Proposition 2. The optimal refinancing threshold $\theta(m)$ is increasing in the attention rate λ .

Interpretation: Less attentive borrowers optimally refinance at lower rate gaps when they do pay attention, because they may not observe another good opportunity soon.

3.5 Brownian mortgage-rate case

When m_t is Brownian with volatility σ , the rate gap $z = c - m$ is sufficient. The value function can be written as

$$V(m, c) = \frac{c}{\rho} + v(z).$$

The paper derives a closed-form threshold θ involving the Lambert W function and constants

$$\eta_0 = \frac{\sqrt{2(\rho + \nu)}}{\sigma}, \quad \eta_\lambda = \frac{\sqrt{2(\rho + \nu + \lambda)}}{\sigma},$$

$$\epsilon_\lambda = \frac{(\rho + \nu)(\eta_0 + \eta_\lambda)}{\lambda}.$$

Two limiting cases are especially useful:

$$\lim_{\lambda \rightarrow 0} \theta = (\rho + \nu)\psi, \quad (3)$$

$$\lim_{\lambda \rightarrow \infty} \theta = \text{ADL perfect-attention threshold}. \quad (4)$$

Interpretation: The ADL formula is the infinite-attention limit. Inattention lowers the threshold substantially relative to this benchmark.

3.6 Distribution of rate gaps

The paper derives ergodic densities for the rate gap z_t and for rate gaps upon prepayment. The key sufficient statistic is

$$\text{avg. gap upon prepay} - \text{avg. gap} = \frac{\sigma}{\sqrt{2}} \left(\frac{1}{\sqrt{\nu}} - \frac{1}{\sqrt{\nu + \lambda}} \right). \quad (5)$$

Interpretation: The threshold θ drops out of the moment difference. Therefore, the attention rate can be recovered from simple moments without knowing fixed costs or discount rates.

3.7 Empirical implementation

Given average gap 0.15%, average gap upon prepayment 0.78%, and $\nu = 11.33\%$, the paper inverts the sufficient statistic:

- if $\sigma = 0.70\%$, $\lambda = 23.4\%$ per year;
- if $\sigma = 1.00\%$, $\lambda = 11.8\%$ per year.

Interpretation: With $\lambda = 23.4\%$, households check mortgage refinancing opportunities roughly once every 4.3 years on average.

4 Identification and empirical design

4.1 What identifies inattention

- The empirical object is not a treatment effect but a structural friction.
- The model maps the difference between average rate gaps and average rate gaps upon prepayment into λ .
- This mapping relies on Brownian rate dynamics, prepayment rates, volatility, and the long-run rate-gap distribution.

Interpretation: The paper identifies inattention by asking what attention rate is needed for the model to produce the observed spread between ordinary rate gaps and prepayment rate gaps.

4.2 Why the model changes mistake measurement

- Under perfect attention, borrowers should refinance immediately after the rate gap exceeds the ADL threshold.
- Under inattention, borrowers refinance only when an attention opportunity arrives.
- Therefore, delayed refinancing above the threshold is not necessarily a mistake.
- Also, the inattention threshold is lower, so some apparently premature refinances are actually optimal.

Interpretation: Many borrower “mistakes” are defined relative to the wrong benchmark if borrowers are inattentive.

4.3 Model extensions

- The main threshold comparative static also holds under a broader class of mortgage-rate processes.
- Mean-reverting mortgage rates produce similar quantitative conclusions.
- A rational-inattention microfoundation with observation costs generates the same core logic: higher effective attention raises the threshold.

Interpretation: The key mechanism is not tied narrowly to Brownian rates or exogenous Poisson attention.

4.4 Limits of the design

- The analytical sufficient statistic depends on the Brownian rate assumption.
- Rate gaps in the data require estimating counterfactual refinancing rates.
- Mortgage prepayments may occur for reasons other than refinancing.

- The attention rate is an average friction across heterogeneous households.
- The model abstracts from many institutional constraints, including employment checks and refinancing frictions beyond fixed costs.

Interpretation: The quantitative results should be read as disciplined structural measurement, not direct observation of household attention.

5 Tables and figures

The paper contains two figures and one main table. Because the paper is theoretical, this section also includes directly cropped original equation panels.

5.1 Equation panel: HJB and attention comparative static

Read:

- The HJB combines interest payments, exogenous mortgage exit, refinancing opportunities, and mortgage-rate dynamics.
- Proposition 2 states that the threshold increases with attention λ .
- Therefore, lower attention means lower optimal thresholds at observed attention times.

Economic message: Inattention does not only delay refinancing; it changes what optimal refinancing means when the borrower finally looks.

$$(3) \quad (\rho + \nu + \lambda)V(m, c) = c + \nu V(m, m) + \lambda \min \{V(m, c), V(m, m) + \psi\} + \mathcal{L}V(m, c).$$

If there exists a function V twice differentiable in m that satisfies (3), then V must be the solution to problem (1).⁷ In our next proposition (proven in Appendix A.A2), we study the extent to which the threshold $\theta(m)$ varies with the degree of borrower inattention.

PROPOSITION 2: *Assume that there exists a function V twice differentiable in m and once differentiable in the parameter λ that satisfies (3). The threshold $\theta(m)$ is an increasing function of λ .*

5.2 Equation panel: Brownian threshold formula

Read:

- The Brownian case yields an explicit threshold formula with the Lambert W function.
- As $\lambda \rightarrow 0$, the threshold converges to the flow value of the fixed cost, $(\rho + \nu)\psi$.
- As $\lambda \rightarrow \infty$, the threshold converges to the ADL perfect-attention threshold.
- The approximation $\hat{\theta}$ gives a tractable lower bound.

Economic message: The formula makes clear why perfect attention overstates the rate gap needed for an inattentive borrower to refinance.

The constants of integration k_-, k_+ are given in Appendix A.A3. The (state-independent) rate gap threshold $\theta > 0$ is given by

$$(5) \quad \theta = (\rho + \nu)\psi + \frac{1}{\eta_0 + \epsilon_\lambda} + \frac{1}{\eta_0} W\left(\frac{-\eta_0}{\eta_0 + \epsilon_\lambda} \exp\left\{\frac{-\eta_0}{\eta_0 + \epsilon_\lambda} [1 + (\rho + \nu)(\eta_0 + \epsilon_\lambda)\psi]\right\}\right),$$

where W is the Lambert W function. θ increases with the attention rate λ , and asymptotically

$$(6) \quad \lim_{\lambda \rightarrow 0} \theta = (\rho + \nu)\psi$$

$$(7) \quad \lim_{\lambda \rightarrow +\infty} \theta = (\rho + \nu)\psi + \frac{1}{\eta_0} + \frac{1}{\eta_0} W\left(-\exp\{-[1 + (\rho + \nu)\eta_0\psi]\}\right).$$

A Taylor expansion of the implicit equation underlying (5) around $\theta = 0$ yields an approximation⁸ and lower bound $\hat{\theta}$ of the threshold θ with formula

$$(8) \quad \hat{\theta} = \sqrt{\frac{2}{\eta_0} \left(1 + \frac{\epsilon_\lambda}{\eta_0}\right) (\rho + \nu)\psi + \left(\frac{\epsilon_\lambda}{\eta_0}\right)^2} - \frac{\epsilon_\lambda}{\eta_0^2}.$$

5.3 Figure 1: Rate gap threshold θ

Read:

- Panel A plots the refinancing threshold as a function of the attention rate λ .
- The threshold rises with attention and converges toward the ADL benchmark.
- For low attention rates, the threshold is much smaller.
- Panel B shows the same logic in the rational-inattention model with observation costs.

Economic message: The figure is the core comparative static: inattentive households should refinance earlier, conditional on noticing.

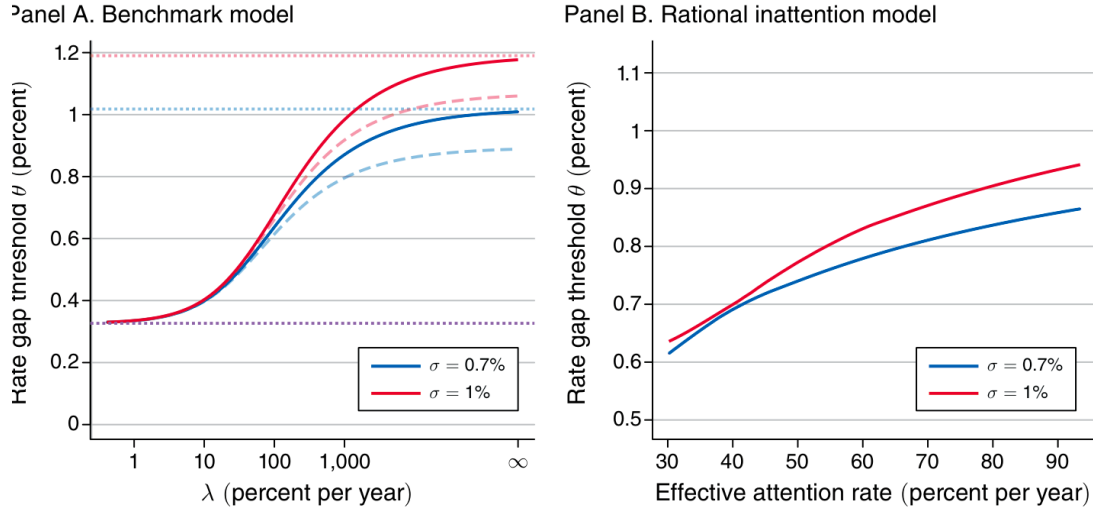


FIGURE 1. RATE GAP THRESHOLD θ

5.4 Equation panel and Figure 2: Distributional implications

Read:

- The rate-gap distribution is asymmetric Laplace under Brownian rates.
- The rate gap upon prepayment has a related but shifted distribution.
- The difference between average gap upon prepayment and average gap yields a sufficient statistic for attention.
- Figure 2 compares theory-implied densities with empirical rate-gap densities.

Economic message: The model can be taken to data using only observable rate-gap moments, not the full value function.

ergodic density $f(z)$.

$$(10) \quad f(z)dz := \lim_{t \rightarrow +\infty} \frac{1}{t} \int_0^t \mathbf{1}_{\{z \in [z, z+dz]\}} dt.$$

The density f can be characterized analytically, as the next proposition shows.

PROPOSITION 4: Assume that m_t is a Brownian motion with volatility σ . Introduce the constants χ_0 and χ_λ equal to

$$\chi_0 := \frac{\sqrt{2\nu}}{\sigma}, \quad \chi_\lambda := \frac{\sqrt{2(\nu + \lambda)}}{\sigma}.$$

The rate gap z_t admits an ergodic density f that is an asymmetric Laplace distribution:

$$(11) \quad f(z) = \frac{1}{\frac{1}{\chi_0} + \frac{1}{\chi_\lambda}} \begin{cases} \exp(\chi_0(z - \theta)), & \text{if } z \leq \theta \\ \exp(-\chi_\lambda(z - \theta)), & \text{if } z \geq \theta \end{cases}$$

Instead, the rate gap upon prepayment admits an ergodic density $\hat{f}$ equal to

$$(12) \quad \hat{f}(z) = \frac{1}{\left(\frac{1}{\chi_0} + \frac{1}{\chi_\lambda}\right)\nu + \frac{\lambda}{\chi_\lambda}} \begin{cases} \nu \exp(\chi_0(z - \theta)), & \text{if } z \leq \theta \\ (\nu + \lambda) \exp(-\chi_\lambda(z - \theta)), & \text{if } z \geq \theta \end{cases}$$

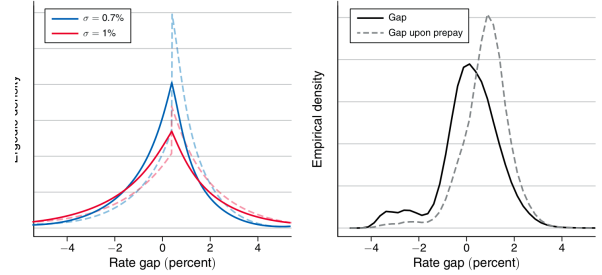
The following ergodic statistics admit analytic expressions:

$$(13) \quad \text{avg. prepay rate} = \lim_{t \rightarrow +\infty} \frac{1}{t} \left(N_t^{(\nu)} + \int_0^t a_s dN_s^{(\lambda)} \right) = \nu + \left(\frac{\chi_0}{\chi_0 + \chi_\lambda} \right) \lambda;$$

$$(14) \quad \text{avg. gap} = \lim_{t \rightarrow +\infty} \frac{1}{t} \int_0^t z_s ds = \theta + \frac{1}{\chi_\lambda} - \frac{1}{\chi_0};$$

$$(15) \quad \text{avg. gap at prepay} = \lim_{t \rightarrow +\infty} \frac{\int_0^t z_s - \left(dN_s^{(\nu)} + a_s dN_s^{(\lambda)} \right)}{t} = \theta.$$

(a) Ergodic distributions and moments.



(b) Theory versus data densities.

5.5 Sufficient statistic equation

Read:

- Equation (16) links the data moments directly to λ .
- The only required model inputs are σ and ν .
- The statistic does not depend on fixed costs ψ or the discount rate ρ .

Economic message: This is the paper's empirical bridge from rate-gap distributions to inattention frictions.

The statistics (14) and (15) are measurable in the data. Since they both increase one-for-one with θ , their difference is independent of this threshold and only depends on the moving rate ν , the attention rate λ , and mortgage rate volatility σ :

5.6 Table 1: Errors of omission and commission

Read:

- With $\sigma = 0.70\%$, the average threshold falls from 1.09% under perfect attention to 0.75% with inattention.
- Under perfect attention, the theoretical monthly prepayment rate above threshold is 100%, but the empirical rate is only 2.97%.
- With inattention, the theoretical monthly prepayment rate above threshold is 2.85%, close to the empirical rate of 2.77%.
- The number of apparent commission errors falls because many low-gap refinances are above the lower inattention threshold.
- For $\sigma = 1.00\%$, the same qualitative conclusions hold.

Economic message: Inattention nearly eliminates errors of omission and rationalizes a large share of errors of commission.

TABLE 1—ERRORS OF OMISSION AND OF COMMISSION

	Perfect attention	With inattention	Perfect attention	With inattention
Mortgage rate (annual) volatility σ	0.70%	0.70%	1.00%	1.00%
Average threshold θ_H	1.09%	0.75%	1.27%	0.71%
<i>Errors of omission</i>				
Mortgage-month observations (M) with $z_H > \theta_H$	7.07	10.87	5.43	11.47
Number of prepayments (M) when $z_H > \theta_H$	0.21	0.3	0.16	0.31
Empirical monthly prepay rate when $z_H > \theta_H$	2.97	2.77	3.01	2.72
Theoretical monthly prepay rate when $z_H > \theta_H$	100	2.85	100	1.91
<i>Errors of commission</i>				
Mortgage-month observations (M) with $z_H < \theta_H$	28.35	24.56	29.99	23.95
Number of prepayments (M) when $z_H < \theta_H$	0.31	0.21	0.35	0.2
Empirical monthly prepay rate when $z_H < \theta_H$	1.08	0.87	1.17	0.85
Theoretical monthly prepay rate when $z_H < \theta_H$	0.94	0.94	0.94	0.94
Mortgage-month observations (M) with $0 < z_H < \theta_H$	13.67	9.88	15.31	9.27
Number of prepayments (M) when $0 < z_H < \theta_H$	0.21	0.12	0.26	0.11
Empirical monthly prepay rate when $0 < z_H < \theta_H$	1.57	1.24	1.7	1.2
Mortgage-month observations (M) with $z_H < 0$	14.68	14.68	14.68	14.68
Number of prepayments (M) when $z_H < 0$	0.09	0.09	0.09	0.09
Empirical monthly prepay rate when $z_H < 0$	0.62	0.62	0.62	0.62

Notes: Random sample of 780,000 mortgages issued between January 1999 and September 2023 (mortgage rate volatility over that sample period is estimated to be equal to 0.70 percent per year). Panel consists of 35.42 million mortgage-month observations. Prepayment rates are expressed in percent per month.

6 Short summary

- The paper studies optimal fixed-rate mortgage refinancing with fixed costs and inattention.
- Borrowers pay attention randomly, according to a Poisson process with intensity λ .
- The optimal policy is a rate-gap threshold rule at attention times.
- Greater attention raises the refinancing threshold; lower attention lowers it.
- Under Brownian mortgage rates, the paper derives closed-form formulas for the refinancing threshold.
- The ADL threshold is recovered as the perfect-attention limit $\lambda \rightarrow \infty$.
- The model implies analytical long-run distributions of rate gaps and rate gaps upon prepayment.
- The difference between average gap upon prepayment and average gap is a sufficient statistic for attention.
- Using Fannie Mae SFLP data, the preferred attention estimate is 23.4% per year.
- With this attention rate, average thresholds are much lower than under perfect attention.
- The model nearly eliminates errors of omission and reduces errors of commission by 32–43 percent.

7 Conclusion

7.1 Core conclusion

Berger, Milbradt, Tourre, and Vavra show that mortgage refinancing behavior looks much less mistaken once inattention is built into the optimal benchmark. Inattention both delays refinancing and lowers the optimal rate-gap threshold at the moment the borrower pays attention.

7.2 Economic intuition

A perfectly attentive borrower can wait for a large rate gap because the borrower will continuously observe future opportunities. An inattentive borrower cannot. When such a borrower finally checks rates, the option value of waiting is lower because another observation may not arrive for years. Thus, the borrower should refinance at a smaller gap.

7.3 Takeaway

The paper changes the benchmark for judging mortgage refinancing mistakes. Many households who seem too slow or too early under perfect attention are behaving in a way that is close to optimal under sporadic attention. The main policy and empirical lesson is that refinancing frictions should be measured relative to an inattentive benchmark, not only the ADL perfect-attention benchmark.